

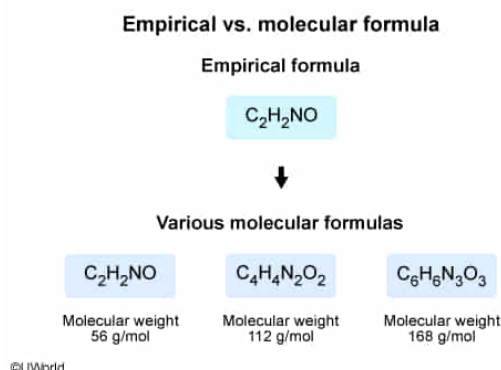
An 84-mg sample of a compound is found to contain 36 mg of carbon, 3 mg of hydrogen, 21 mg of nitrogen, and 24 mg of oxygen. If the compound has a molecular weight of 112 g/mol, what is its molecular formula?

- ☐ A. $\text{C}_2\text{H}_2\text{NO}$ [14%]
☐ B. $\text{C}_4\text{H}_6\text{N}_3\text{O}$ [16%]
☐ C. $\text{C}_5\text{H}_6\text{NO}_2$ [15%]
✔ ☒ D. $\text{C}_4\text{H}_4\text{N}_2\text{O}_2$ [54%]

Correct

54% answered correctly

Explanation:



The composition of chemical compounds can be described in several ways, including the empirical, molecular, and structural formulae. The **empirical formula** describes only the **ratios** of each unique atom in a compound, whereas the **molecular formula** describes the **actual number** of each type of atom. The **structural formula** describes **how the atoms are bonded** to one another.

An empirical formula may be calculated from the known mass of each type of atom in a compound sample by calculating the number of moles of each. Dividing each mass given in the question by the **atomic mass of the atom** yields the number of moles. For these calculations, grams and moles are scaled down to milligrams and millimoles:

- Carbon: $\frac{36 \text{ mg}}{12 \text{ mg/mmol}} = 3 \text{ mmol}$
- Hydrogen: $\frac{3 \text{ mg}}{1 \text{ mg/mmol}} = 3 \text{ mmol}$
- Nitrogen: $\frac{21 \text{ mg}}{14 \text{ mg/mmol}} = 1.5 \text{ mmol}$
- Oxygen: $\frac{24 \text{ mg}}{16 \text{ mg/mmol}} = 1.5 \text{ mmol}$

Dividing each molar amount by the smallest molar amount (1.5 mmol) yields the relative ratios of each type of atom: C:H:N:O = 2:2:1:1. Therefore, the empirical formula of the compound

is $\text{C}_2\text{H}_2\text{NO}$ (**Choice A**). However, the question states that the compound has a molecular weight of 112 g/mol. A compound with a molecular formula of $\text{C}_2\text{H}_2\text{NO}$ would only have a molecular weight of 56 g/mol, or half the observed molecular weight. Doubling the number of each atom (a two-fold multiple of the lowest whole-number ratio) yields the correct molecular formula $\text{C}_4\text{H}_4\text{N}_2\text{O}_2$, which is consistent with the compound composition.

(Choices B and C) $\text{C}_4\text{H}_6\text{N}_3\text{O}$ and $\text{C}_5\text{H}_6\text{NO}_2$ have the correct molecular weight but the wrong ratios of each type of atom.

Educational objective:

Compounds may be described by their empirical, molecular, and structural formulae. Empirical formulae describe only the relative ratios of each type of atom, molecular formulae describe the actual numbers of each type of atom, and structural formulae describe the spatial arrangement of the atoms.

Time spent: 0 sec

Subject:
General Chemistry

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System:
Atomic Theory and Chemical Composition